



Advanced Time Series Analysis and Forecasting

Seminar 2: Structural breaks and nonlinear models



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Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BIC:** Bayesian Information Criterion
- ▣ **BIS:** Bank for International Settlements
- ▣ **BNR:** Banca Națională a României (the National Bank of Romania)
- ▣ **COVID:** COronaVirus Disease (2019 pandemic)
- ▣ **CPI:** Consumer Price Index
- ▣ **CRPS:** Continuous Ranked Probability Score
- ▣ **CSW:** Chu, Stinchcombe and White (monitoring boundary)
- ▣ **DF:** Dickey–Fuller (test)
- ▣ **DM:** Diebold–Mariano (test)
- ▣ **EODHD:** EOD Historical Data (market-data provider)
- ▣ **ESTAR:** Exponential Smooth Transition AutoRegression
- ▣ **FRED:** Federal Reserve Economic Data
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP:** Gross Domestic Product
- ▣ **GW:** Giacomini–White (test of conditional predictive ability)

Acronyms used in this material (2/2)

- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ **ICSS**: Iterated Cumulative Sums of Squares
- ▣ **IT**: Inclán–Tiao (statistic of the ICSS algorithm)
- ▣ **KSS**: Kapetanios, Shin and Snell (unit-root test against ESTAR)
- ▣ **LM3**: Lagrange Multiplier linearity test, third-order expansion
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **LSTAR**: Logistic Smooth Transition AutoRegression
- ▣ **LWZ**: Liu, Wu and Zidek (modified Schwarz criterion)
- ▣ **MSFE**: Mean Squared Forecast Error
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **SCA**: Seasonally and Calendar Adjusted
- ▣ **SETAR**: Self-Exciting Threshold AutoRegression
- ▣ **SSR**: Sum of Squared Residuals
- ▣ **STAR**: Smooth Transition AutoRegression
- ▣ **TAR**: Threshold AutoRegression
- ▣ **US**: United States

Today's question and route

- **Question:** how do we date breaks and test for nonlinearity when the break date or the threshold is not known in advance?
 - ▶ this seminar comes **before** Lecture 2: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: the Chow test and the Wald process; dynamic programming and the choice of the number of breaks; a SETAR skeleton and the threshold LR; ESTAR half-lives; monitoring and the κ_2 correction; the optimal window
 - ▶ Part B: Romanian GDP volatility; the Romanian 3-month rate; sunspots; the Romanian real effective exchange rate; BET volatility
 - ▶ Part C: the estimation window for Romanian inflation forecasts; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	the Chow test; the Wald process of a mean shift	Solved, Proposed	A1
A3, A4	dynamic programming by hand; BIC, LWZ and sequential tests	Solved, Proposed	A3
A5, A6	a SETAR skeleton and the threshold LR; ESTAR persistence	Solved, Proposed	A5
A7, A8	the monitoring boundary and κ_2 ; the optimal window	Solved, Proposed	A7
B1, B2	Romanian GDP volatility; Bai–Perron on the 3-month rate	Solved, Proposed	B1
B3–B5	SETAR for sunspots; ESTAR for the RON real exchange rate; BET variance breaks	Solved, Proposed	B1, B3
C1, C2	window choice for inflation forecasts; what is wrong in an AI answer?	Proposed	A8, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
real GDP, Romania (SCA)	Eurostat (namq_10_gdp)	quarterly	1995–2026
3-month money-market rate, Romania	Eurostat (irt_st_m)	monthly	2005–2026
sunspot numbers	statsmodels data set	yearly	1700–2008
real effective exchange rate, Romania	BIS via FRED (RBROBIS)	monthly	1994–2026
BET	EODHD	daily	2000–2026
HICP inflation, Romania	Eurostat (prc_hicp_minr)	monthly	2000–2026

- In the notebook: `andrews_tests`, `bp_analysis`, `tar_estimate`, `hansen_test`, `lm3_test`, `estar_fit`, `icss`, `csw_boundary`; no account or key is needed
- BIS: Bank for International Settlements; SCA: seasonally and calendar adjusted

Prerequisites for Today (1/6): one break at a known date

- Model $y_t = x_t' \beta_t + u_t$, $t = 1, \dots, T$, with k regressors x_t and coefficients β_t
 - ▶ one break at $T_1 = [\pi T]$: $\beta_t = \beta_1$ before, β_2 after; $\pi \in (0, 1)$: the break fraction; $[\cdot]$: the integer part
- **Chow** test [Chow \(1960\)](#): SSR (sum of squared residuals) of the pooled fit S_0 and of the two sub-samples S_1 , S_2
 - ▶ $F = \frac{(S_0 - S_1 - S_2)/k}{(S_1 + S_2)/(T - 2k)} \sim F(k, T - 2k)$
 - ▶ the numerator is the gain in fit per extra coefficient; a large F is evidence of a break
- **Predictive** Chow test, $n_2 = T - T_1$ new observations:
 - ▶ $F = \frac{(S_0 - S_1)/n_2}{S_1/(T_1 - k)} \sim F(n_2, T_1 - k)$
 - ▶ does the model estimated up to T_1 still fit the new observations?

Prerequisites for Today (2/6): one break at an unknown date

- Unknown date: take the largest Wald statistic over the candidate fractions (sup-Wald) [Andrews \(1993\)](#)
 - ▶ $\sup_{\pi \in [0.15, 0.85]} W_T(\pi)$
 - ▶ $W_T(\pi)$: the Wald statistic of no break at fraction π ; 15% trimming at each end; exp- and ave-Wald average $W_T(\pi)$ instead [Andrews and Ploberger \(1994\)](#)
 - ▶ the search raises the critical value: 5% value of the supremum (one coefficient): 8.65, against 3.84 at a known date
- Mean shift with known variance σ^2 :
 - ▶ $W_T(\pi) = T\pi(1 - \pi)(\bar{y}_1 - \bar{y}_2)^2 / \sigma^2$
 - ▶ $\bar{y}_1, \bar{y}_2$: the means before and after $[\pi T]$
- Variance break [McConnell and Perez-Quiros \(2000\)](#): apply the test to $\sqrt{\pi/2} |\hat{e}_t|$
 - ▶ $\hat{e}_t$: the residuals; here $\pi = 3.14 \dots$; under Normality $\mathbb{E}\sqrt{\pi/2} |e_t| = \sigma$, so a break in this mean is a break in the standard deviation

Prerequisites for Today (3/6): several breaks

- **Bai–Perron** Bai and Perron (1998), Bai and Perron (2003): minimise the total SSR over all partitions with segments of at least h observations
 - ▶ $SSR_r(n) = \min_j [SSR_{r-1}(j) + SSR(j+1, n)]$
 - ▶ $SSR_r(n)$: the SSR of the best r -break partition of the first n observations; $SSR(i, n)$: of one regression on $i, \dots, n$; j : the last break
 - ▶ tests: $\sup F(k)$ (0 against k breaks), $UD_{\max} = \max_k \sup F(k)$, sequential $F(\ell+1 \mid \ell)$ (add a break while it rejects)
- Criteria with $p^* = 2m + 1$ parameters for m mean shifts ($m+1$ means and m dates)
 - ▶ $BIC(m) = \ln(SSR_m/T) + p^* \ln T/T$; $LWZ(m) = \ln[SSR_m/(T - p^*)] + p^* \cdot 0.299(\ln T)^{2.1}/T$ Liu, Wu and Zidek (1997)
 - ▶ choose the m with the smallest value; LWZ penalises extra breaks more than BIC

Prerequisites for Today (4/6): dates, monitoring, variance

- Confidence interval for a date [Bai \(1997\)](#), with a mean shift of size δ :
 - ▶ $(\delta^2/\sigma^2)(\hat{T}_1 - T_1) \rightarrow_d \arg \max_s [W(s) - |s|/2]$
 - ▶ W : a two-sided Brownian motion; with equal variances the 2.5% and 97.5% quantiles are about ∓ 11 : half-width $\approx 11\sigma^2/\delta^2$ periods
- **Monitoring** [Chu, Stinchcombe and White \(1996\)](#): after m historical observations, alarm at the first n with
 - ▶ $|\sum_{t=m+1}^n w_t/\hat{\sigma}| > \sqrt{n[a^2 + \ln(n/m)]}$
 - ▶ w_t : recursive residuals (one-step forecast errors, standardised); a : a constant; size over an infinite horizon $2[1 - \Phi(a) + a\varphi(a)]$ [Robbins and Siegmund \(1970\)](#)
- **Variance breaks** [Inclán and Tiao \(1994\)](#): a_t demeaned returns, $C_k = \sum_{t \leq k} a_t^2$, $D_k = C_k/C_T - k/T$
 - ▶ $IT = \sqrt{T/2} \max_k |D_k|$, 5% value 1.358; the factor assumes $\text{Var}(a_t^2) = 2\sigma^4$ (Normal data)
 - ▶ κ_2 [Sansó, Aragón and Carrion \(2004\)](#) uses a long-run variance of a_t^2 instead; with kurtosis κ (i.i.d.) IT is too large by a factor $\sqrt{(\kappa - 1)/2}$

Prerequisites for Today (5/6): windows and threshold models

- **Window** after a mean shift δ , n_2 post-break observations, forecast = mean of the last $w \geq n_2$ **Pesaran and Timmermann (2007)**
 - ▶ $\text{MSFE}(w) = \sigma^2(1 + 1/w) + [(w - n_2)\delta/w]^2$
 - ▶ MSFE: the mean squared forecast error; variance falls and bias grows with w ; AveW averages the forecasts of all windows **Pesaran and Pick (2011)**
- **SETAR** **Tong and Lim (1980)**: an AR whose coefficients depend on whether y_{t-d} is below or above γ
 - ▶ $y_t = (\phi_{1,0} + \phi'_1 y_{t-1})1\{y_{t-d} \leq \gamma\} + (\phi_{2,0} + \phi'_2 y_{t-1})1\{y_{t-d} > \gamma\} + \varepsilon_t$
 - ▶ y_{t-1} : the last p values; $\phi_{j,0}$, ϕ_j : intercept and AR coefficients of regime j ; d : the delay lag; γ : the threshold
 - ▶ skeleton: the map with $\varepsilon_t = 0$; SETAR(2; 1, 1) is ergodic iff $\phi_1 < 1$, $\phi_2 < 1$, $\phi_1\phi_2 < 1$
- Inference on the threshold
 - ▶ linearity test: $\sup_{\gamma} W(\gamma)$, p -value by the fixed-regressor bootstrap $y_t^* = \hat{e}_t \eta_t$, $\eta_t \sim N(0, 1)$ **Hansen (1996)**
 - ▶ threshold set $\{\gamma : \text{LR}(\gamma) \leq c(\alpha)\}$, $c(\alpha) = -2\ln(1 - \sqrt{1 - \alpha})$; LR: the relative rise of the SSR away from $\hat{\gamma}$ **Hansen (2000)**

Prerequisites for Today (6/6): smooth transition

- **STAR Teräsvirta (1994)**: two AR regimes mixed by a weight $G(s_t; \gamma, c) \in [0, 1]$
 - ▶ s_t : the transition variable; c : the location; γ : the speed; logistic G (LSTAR) or $G = 1 - e^{-\gamma(s-c)^2}$ (ESTAR)
 - ▶ LM3 test from a Taylor expansion of G ; sequence H_{04}, H_{03}, H_{02} : the smallest p -value for H_{03} points to ESTAR
- **ESTAR of Taylor, Peel and Sarno (2001)** for the real exchange rate q_t , equilibrium μ :
 - ▶ $q_t - \mu = (q_{t-1} - \mu)e^{-\theta^2(q_{t-1} - \mu)^2} + \varepsilon_t$
 - ▶ local AR coefficient $\rho(x) = e^{-\theta^2 x^2}$ at deviation x : close to 1 near μ , small far away; half-life $\ln 0.5 / \ln \rho$
 - ▶ KSS test **Kapetanios, Shin and Snell (2003)**: t statistic of δ in $\Delta q_t = \delta q_{t-1}^3 + e_t$ (demeaned q); reject a unit root in the left tail

A1: the Chow test by hand [Solved]

Task

- ▣ A regression with $k = 2$ coefficients on $T = 80$ quarters; break candidate after observation 40. SSR: pooled 120, first half 45, second half 50.
- ▣ 1. Compute the Chow statistic and its 5% critical value; conclude.
- ▣ 2. Six new quarters arrive; the SSR of the first 40 plus the six new ones is 52. Compute the predictive Chow statistic.
- ▣ 3. The date 40 was chosen after looking at a plot. Which critical value should replace the F one?
- ▣ Report: two statistics, two critical values, one sentence.

Solution

- ▣ 1. $F = \frac{(120 - 95)/2}{95/76} = 10.00$; $F_{0.95}(2, 76) = 3.12$, $p < 0.001$: the coefficients change
- ▣ 2. $F = \frac{(52 - 45)/6}{45/38} = 0.985$, $F_{0.95}(6, 38) = 2.35$, $p = 0.449$: the model still forecasts the new quarters
- ▣ 3. the sup-Wald value for $p = 2$ restrictions (about 11.7 at 5% with 15% trimming); the Wald form is $2F = 20$: still a rejection

A2: the Wald process of a mean shift [Proposed]

Solution discussed in class

Task

- ▣ $y_t = \mu + \delta \mathbf{1}\{t > [\pi T]\} + u_t$, u_t i.i.d. $N(0, \sigma^2)$, $\sigma^2 = 1$ known, $T = 100$. Model: A1.
- ▣ 1. Show that the Wald statistic for $\delta = 0$ at fraction π is $W_T(\pi) = T\pi(1 - \pi)(\bar{y}_1 - \bar{y}_2)^2 / \sigma^2$.
- ▣ 2. The largest Wald statistic is found at $\pi = 0.3$ with $\bar{y}_1 - \bar{y}_2 = 0.5$. Compute it and compare with $\chi_{0.95}^2(1)$ and with the sup-Wald value.
- ▣ 3. Explain why the search range must exclude π close to 0 and 1.
- ▣ Report: one derivation, one statistic, one decision, one sentence.

A3: dynamic programming on eight observations [Solved]

Task

- ▣ $y = (1, 1, 2, 1, 5, 6, 5, 6)$, mean-shift model, minimum segment length $h = 2$.
- ▣ 1. Compute $SSR(i, j)$ for the segments needed by the one-break problem and find the best single break.
- ▣ 2. Use the recursion to find the best two-break partition.
- ▣ 3. How much does the second break reduce the SSR?
- ▣ Report: the SSR table, two partitions, one comparison.

Solution

- ▣ 1. no break: SSR 37.875; candidates: $[1, 2] + [3, 8]: 0.000 + 22.833$; $[1, 3] + [4, 8]: 0.667 + 17.200$; $[1, 4] + [5, 8]: 0.750 + 1.000 = 1.75$; $[1, 5] + [6, 8]: 12.000 + 0.667$; $[1, 6] + [7, 8]: 25.333 + 0.500$
- ▣ best break after observation 4 (means 1.25 and 5.5)
- ▣ 2. $SSR_2(8) = \min_j [SSR_1(j) + SSR(j+1, 8)]$; the minimum is $[1, 2], [3, 4], [5, 8]: 0 + 0.5 + 1 = 1.50$
- ▣ 3. from 1.75 to 1.50: a small gain that BIC would not buy; the first break explains almost everything

A4: BIC, LWZ and the sequential tests [Proposed]

Solution discussed in class

Task

- ▣ The US real rate of the lecture: $T = 103$, SSR for $m = 0, 1, 2, 3$ mean shifts: 1214.9, 645.0, 456.0, 445.2. Model: A3.
- ▣ 1. Compute $\text{BIC}(m)$ and $\text{LWZ}(m)$ for $m = 0, \dots, 3$ with $p^* = 2m + 1$ and choose m by each criterion.
- ▣ 2. The sequential statistics are $F(2|1) = 36.86$, $F(3|2) = 16.09$, $F(4|3) = 0.04$ with 5% values 10.54, 11.56, 11.96. How many breaks does the sequential procedure select?
- ▣ 3. Explain why the two approaches can disagree.
- ▣ Report: a table of eight values, two choices, one sentence.

A5: a SETAR skeleton and the threshold LR [Solved]

Task

- ▣ SETAR(2; 1, 1) with $d = 1$, threshold 0: $y_t = 0.4 + 0.5y_{t-1}$ if $y_{t-1} \leq 0$; $y_t = -0.6 + 0.8y_{t-1}$ if $y_{t-1} > 0$ (plus noise).
- ▣ 1. Is the process ergodic? Compute the regime means $\phi_{j,0}/(1 - \phi_j)$ and say whether they lie in their own regimes.
- ▣ 2. Iterate the skeleton from $y_0 = 1$ for five steps and find its limit.
- ▣ 3. Derive the 95% and 90% critical values of the threshold LR from $P(\xi \leq x) = (1 - e^{-x/2})^2$.
- ▣ Report: one condition, five numbers, a limit, two critical values.

Solution

- ▣ 1. $\phi_1 = 0.5 < 1$, $\phi_2 = 0.8 < 1$, $\phi_1\phi_2 = 0.4 < 1$: ergodic; regime means 0.8 (regime 1 needs $y \leq 0$) and -3.0 (regime 2 needs $y > 0$): both are *virtual*
- ▣ 2. 0.200, -0.440 , 0.180, -0.456 , 0.172: a two-cycle
 $a = (0.4 - 0.5 \cdot 0.6)/(1 - 0.4) = 0.167$,
 $b = -0.6 + 0.8a = -0.467$
- ▣ 3. $(1 - e^{-x/2})^2 = 1 - \alpha \Rightarrow x = -2 \ln(1 - \sqrt{1 - \alpha})$: 7.35 (95%) and 5.94 (90%)
- ▣ a nonlinear model can oscillate forever without noise; no linear AR(1) can

A6: local persistence in an ESTAR [Proposed]

Solution discussed in class

Task

- ▣ The dollar–sterling ESTAR of Taylor, Peel and Sarno (2001): $\theta^2 = 0.452$; a linear AR(1) fitted to the same data has a half-life of about 45 months. Model: A5.
- ▣ 1. Compute the local AR coefficient $\rho(x) = e^{-\theta^2 x^2}$ for deviations $x = 0.05, 0.20, 0.40$ (5%, 20%, 40%).
- ▣ 2. Compute the implied “local” half-lives $\ln 0.5 / \ln \rho(x)$ and the AR(1) coefficient that gives 45 months.
- ▣ 3. Explain why a Dickey–Fuller test has low power when the data follow this ESTAR.
- ▣ Report: three coefficients, four half-lives, one sentence.

A7: the monitoring boundary and the κ_2 correction [Solved]

Task

- ▣ Monitoring with $m = 60$ historical months; ICSS on $T = 500$ returns with $C_{250}/C_{500} = 0.70$.
- ▣ 1. Compute the asymptotic size $2[1 - \Phi(a) + a\varphi(a)]$ for $a^2 = 7.78$ and $a^2 = 6.25$.
- ▣ 2. Compute the 5% boundary $\sqrt{n(a^2 + \ln(n/m))}$ at $n = 90$ and $n = 120$.
- ▣ 3. Compute the IT statistic at $k = 250$; if the returns are i.i.d. with kurtosis 6, compute the corrected statistic.
- ▣ Report: two sizes, two boundaries, two statistics, one decision.

Solution

- ▣ 1. $a = 2.789$: 5.08%; $a = 2.5$: 10.01%
- ▣ 2. $\sqrt{90(7.78 + \ln 1.5)} = 27.14$; $\sqrt{120(7.78 + \ln 2)} = 31.89$: the boundary widens like $\sqrt{n \ln n}$
- ▣ 3. $D_{250} = 0.70 - 0.50 = 0.20$, $IT = \sqrt{250} \cdot 0.20 = 3.16$; factor $\sqrt{(6-1)/2} = 1.58$, corrected $2.00 > 1.358$: still a break, but much weaker evidence

A8: the optimal estimation window [Proposed]

Solution discussed in class

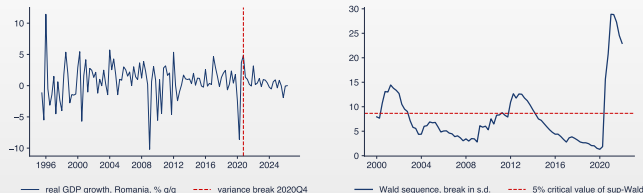
Task

- ▣ A mean shift $\delta = 0.5\sigma$, $\sigma = 1$, happened $n_2 = 20$ periods ago; we forecast with the mean of the last $w \in [20, 200]$ observations. Model: A7.
- ▣ 1. Derive $\text{MSFE}(w) = \sigma^2(1 + 1/w) + [(w - n_2)\delta/w]^2$.
- ▣ 2. Find the optimal w and compare its MSFE with $w = 20$ and $w = 200$.
- ▣ 3. How does the optimal window change with δ ?
- ▣ Report: one derivation, three MSFEs, one sentence.

B1: a Great Moderation in Romania? [Solved]

- **Question:** has the volatility of Romanian GDP growth fallen, and when?
- **Inputs:** real GDP growth q/q (% , Eurostat, SCA), 1995Q3–2026Q2 ($T = 124$); the two-step method of [McConnell and Perez-Quiros \(2000\)](#)
- **Tasks**
 1. Fit an AR(1) and test for a break in its coefficients with sup-Wald (robust variance).
 2. Test for a break in the mean of $\sqrt{\pi/2}|\hat{e}_t|$ with sup-, exp- and ave-Wald (HAC) and date it.
 3. Repeat the variance test without the four quarters of 2020.
 4. Interpretation: is the estimated date a change of regime or the effect of a few quarters?
- **Report:** two sup-Wald statistics with p -values, a date, two standard deviations, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- ▣ AR(1): $\hat{\phi} = -0.05$; break in the coefficients: sup-Wald 3.32 (p 0.838): none
- ▣ Variance: sup 28.9 ($p < 0.001$), exp 10.9, ave 7.9 (5% value of sup: 8.65); date 2020Q4; s.d. 3.00 before, 0.96 after
- ▣ Without 2020: sup 34.0 ($p < 0.001$), date 2012Q4
- ▣ Interpretation: volatility did fall, but the date depends on two COVID quarters; without them the break moves to 2012, after the post-crisis adjustment: report both and let a second break or a robust scale decide

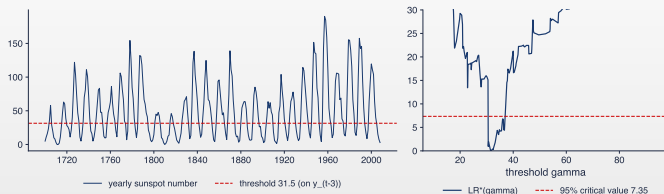
B2: regimes of the Romanian 3-month rate [Proposed]

- **Question:** how many mean regimes does the Romanian money-market rate have since 2005, and do the tests agree?
- **Inputs:** 3-month rate, monthly (Eurostat irt_st_m), January 2005–August 2026 ($T = 260$); mean shifts, $\varepsilon = 0.15$, $M = 5$; model: B1 and A4
- **Tasks**
 1. Estimate the optimal partitions for $m = 1, \dots, 5$ by dynamic programming.
 2. Report $\sup F(1)$, UDmax and the sequential statistics (robust variance) and the number of breaks they select.
 3. Report the numbers of breaks chosen by BIC and LWZ, the dates, the regime means and the 95% intervals.
 4. Interpretation: which partition would you report to the BNR, and why?
- **Report:** three test results, three choices of m , dates with intervals, one sentence
- **Notebook:** section B2

B3: a SETAR for the sunspots [Solved]

- **Question:** is the sunspot cycle better described by a two-regime SETAR than by a linear AR(3)?
- **Inputs:** yearly sunspot numbers 1700–2008 (statsmodels); SETAR(2; 3, 3), $d = 1, \dots, 5$; trimming 15%; a series also studied by [Tong and Lim \(1980\)](#); model: A5
- **Tasks**
 1. Choose the delay d by the smallest SSR over the threshold grid.
 2. Test linearity with the heteroskedasticity-robust sup-Wald and the fixed-regressor bootstrap.
 3. Estimate the threshold and its 95% LR interval; compare the AR(1) and AR(2) coefficients of the two regimes.
 4. Interpretation: what does the threshold separate in the solar cycle?
- **Report:** five SSRs, one bootstrap p -value, a threshold with its interval, one sentence
- **Notebook:** section B3

B3: solution [Solved]



- ▣ SSR for $d = 1, \dots, 5$: 74 620, 58 448, 57 021, 72 363, 74 201: $\hat{d} = 3$; the SETAR reduces the SSR of the AR(3) by 31%
- ▣ sup-Wald 94.1, bootstrap $p < 0.001$ (500 replications); $\hat{\gamma} = 31.5$, 95% set $[30.6; 36.7]$; 123 and 181 years
- ▣ Lower regime: AR(1) 1.91, AR(2) -1.78 (strongly oscillatory); upper regime: 0.86, 0.11
- ▣ Interpretation: the threshold separates the quiet phase (low activity three years earlier), where the next rise is fast and regular, from the active phase, where the decline is slower: the asymmetry of the solar cycle

B4: nonlinear mean reversion of the RON real exchange rate [Proposed]

- **Question:** does the Romanian real effective exchange rate revert to a mean nonlinearly, and how fast?
- **Inputs:** log real effective exchange rate (BIS, CPI-based, via FRED RBROBIS), January 1994–July 2026 ($T = 391$); model: B3 and A6
- **Tasks**
 1. Run the LM3 linearity test with $s_t = q_{t-1}$ and the sequence H_{04}, H_{03}, H_{02} .
 2. Compute the KSS statistic and its simulated 5% value.
 3. Estimate the ESTAR of [Taylor, Peel and Sarno \(2001\)](#) and the half-lives for shocks of 1%, 10% and 40%; compare with a linear AR(1).
 4. Interpretation: is ESTAR the right family for this series?
- **Report:** four p -values, one KSS test, two parameters, four half-lives, one sentence
- **Notebook:** section B4

B5: variance regimes of the BET index [Proposed]

- **Question:** how many breaks in the unconditional variance do BET returns have, once fat tails and volatility clustering are accounted for?
- **Inputs:** daily log returns of the BET index (EODHD), 2000–2026 ($T = 6\,670$); model: A7 and the EUR/RON case of the lecture
- **Tasks**
 1. Run ICSS with the Inclán–Tiao statistic and count the breaks.
 2. Run ICSS with the κ_2 statistic and report the dates and the standard deviation of each regime.
 3. Compare the two whole-sample statistics with the 5% value 1.358.
 4. Interpretation: which set of breaks would you give to a risk manager, and why?
- **Report:** two counts, the κ_2 dates, the regime standard deviations, one sentence
- **Notebook:** section B5

C1: the window for twelve-month inflation forecasts [Proposed]

- **Question:** does a rolling or a window-averaged estimation beat the expanding window for Romanian inflation twelve months ahead, after correcting for three comparisons?
- **Inputs:** HICP inflation, y/y ; direct AR(2) forecasts, targets January 2015–August 2026 ($P = 140$); windows: expanding, rolling 60 and 120 months, AveW (36 months to all); model: A8 and Lecture 2
- **Tasks**
 1. Write the pre-registration (target, horizon, loss, windows, test, correction) in five lines before computing anything.
 2. Compute the RMSE of the four schemes, using at each origin only targets already observed.
 3. Test each scheme against the expanding window with DM–HLN ($h = 12$) and apply the Holm correction (Holm (1979)).
 4. Interpretation: why can the ranking at twelve months differ from the one at one month in the lecture?
- **Report:** a five-line plan, four RMSEs, three HLN statistics with raw and Holm p -values, a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise breaks and threshold models. The answer:
- (a) The 5% critical value of the Andrews sup-Wald test with one restriction and 15% trimming is 3.84.
- (b) The Bai-Perron break fractions converge at rate $\sqrt{T}$, like the regression coefficients.
- (c) A SETAR in which one regime is explosive on its own can never be stationary.
- (d) The 95% confidence set for a threshold collects the values with $LR(\gamma) \leq 3.84$.
- (e) A 5% test repeated at every new observation keeps the overall false-alarm rate at 5%.
- (f) With fat tails the Inclan-Tiao statistic finds too many variance breaks; κ_2 corrects it with the long-run variance of the squared returns.
- (g) In an LSTAR, letting γ tend to 0 gives a threshold model.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ The Chow test is valid only for a date fixed in advance; searched dates need sup-type critical values
- ▣ Dynamic programming gives all optimal partitions; tests and criteria may disagree on their number
- ▣ Threshold models: bootstrap the linearity test, invert LR with 7.35, look at the skeleton
- ▣ ESTAR persistence depends on the size of the deviation; trends and breaks can mimic it
- ▣ An AI answer is a draft: check the critical values, the rates of convergence and the limits of the transition function

After the seminar

- ▣ Lecture 2 develops each topic of today: sup-Wald and its limit, Bai–Perron, monitoring, variance breaks, unit roots with breaks, windows, TAR, STAR and nonlinear forecasting
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: forecasting Romanian inflation across regimes
- ▣ Reading: [Andrews \(1993\)](#); [Bai and Perron \(2003\)](#); [Hansen \(1997\)](#); [Taylor, Peel and Sarno \(2001\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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